

R pbirthday() Function

The pbirthday() Function in R

The `pbirthday()` function calculates probabilities for the generalised birthday paradox.

It computes the probability that, out of a group of items or people, at least a specific number fall into the exact same category (such as sharing a birthday).

Syntax

```
pbirthday(n, classes=365, coincident=2)
```

Parameter	Description	Default Value
<i>n</i>	The number of people (or items) in the group.	<i>Required</i>
<i>classes</i>	The number of available distinct categories (e.g. 365 days in a standard year).	365
<i>coincident</i>	The minimum number of people falling into the exact same category.	2

Example 1 — Standard Birthday Paradox

```
# Probability that among 23 people,  
# there are 2+ people born on the same day  
pbirthday(n = 23, classes = 365, coincident = 2)  
# Output: [1] 0.5072972
```

Example 2 — Birth-hour Problem

```
# Probability that among 1600 people,  
# there are 4+ people who were born not only on the same day,  
# but also at the same hour  
pbirthday(1600, classes = 8760, coincident = 4)  
# Output: [1] 0.296
```